Fiscal Year:	FY 2006	**Task Last Updated:**	FY 08/08/2005
PI Name:	Chen, David J. Ph.D.		
Project Title:	DNA Damage Responses Induced By HZE Particles In Human Cells		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation Biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	75390	**Congressional District:**	30
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Space Radiation Biology and Shielding 03-OBPR-07
Start Date:	10/01/2004	**End Date:**	09/30/2008
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNA05CM04G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	In this newly funded NASA research project (November, 2004), we propose to verify the hypothesis that high energy HZE particles would generate a different and/or unique radiation response in mammalian cells as compared to X-rays. The primary objective of this project is to verify the hypothesis that high energy HZE particles would generate a different and/or unique radiation response in mammalian cells as compared to X-rays. Further more, we will verify whether HZE radiation would induce a bystander effect. Towards this end we will utilize recent advances in our knowledge of molecular mechanisms of DNA damage sensing, signaling, and repair, and changes in global gene expression patterns. Task Objective: 1) To test the hypothesis that HZE particles may induce a different gene expression profile than that induced by low LET radiation; 2) To verify the hypothesis that DNA double-strand breaks induced by HZE particles are persistently		

	unrejoined in normal human fibroblasts; 3) To test the hypothesis that HZE particles would induce a persistent and/or altered molecular and cellular response than that induced by X-rays in mammalian cells; 4) To test the hypothesis that high HZE may generate factors that induce a bystander effect or bystander-like effect. The proposed research should reveal the fundamental differences in cellular responses between low LET X-ray and HZE exposures in humans. These studies would, in turn, provide biological markers for risk assessment and guidance in radiation protection for astronauts during space missions in the future.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Preliminary experiments and feasibility studies have already been carried out at NASA Space Radiation Laboratory (NSRL) at Brookhaven National Laboratory (BNL) during the March 05 NSRL-5 run. Data recently generated from a Jun 05 NSRL-6 run are currently undergoing detailed analyses. We find that 1 GeV Fe, like X-rays, triggers a rapid DNA damage response (visualized as gamma-H2AX phosphorylation and DNA-PK activation). However, the response may be more extensive in response to Fe. Moreover, while the cell is able to repair both X-ray and Fe-induced damage, the kinetics of repair is slower in response to Fe and the residual unrepaired damage levels higher. This could, in turn, account for the higher RBE of Fe as compared to X-rays. We also find evidence for differences in gene expression profiles in response to IR and Fe. Finally, feasibility studies involving the use of CR-39 plastic detectors indicate that this approach will be very useful for studying bystander effects during future NSRL runs.
Bibliography Type:	Description: (Last Updated: 08/14/2018)
Articles in Peer-reviewed Journals	Ding LH, Shingyoji M, Chen F, Hwang JJ, Burma S, Lee C, Cheng JF, Chen DJ. "Gene expression profiles of normal human fibroblasts after exposure to ionizing radiation: a comparative study of low and high doses." Radiat Res. 2005 Jul;164(1):17-26. PMID: 15966761 , Jul-2005